

Tutorial Questions (Set-IX)

Q1 Let $L(\mathbb{R}^n, \mathbb{R}^m)$ be the vector space of linear transformations from $\mathbb{R}^n$ to $\mathbb{R}^m$ and $A \in L(\mathbb{R}^n, \mathbb{R}^m)$. Suppose $\{e_1, e_2, \dots, e_n\}$ and $\{u_1, u_2, \dots, u_m\}$ are standard bases of $\mathbb{R}^n$ and $\mathbb{R}^m$, respectively. Define $\|A\| = \sup\{\|Ax\| : x \in \mathbb{R}^n \text{ with } \|x\| \leq 1\}$.

- (i) Show that $\|A\| \leq \sum_{j=1}^n \|Ae_j\|$. Show that the map given by $A \mapsto \|A\|$ is a norm function on $L(\mathbb{R}^n, \mathbb{R}^m)$. ($\|A\|$ is called the *operator norm* of A .)
- (ii) Show that $\|Ay\| \leq \|A\| \|y\|$ for all $y \in \mathbb{R}^n$. If $\lambda \in \mathbb{R}$ such that $\|Ay\| \leq \lambda \|y\|$ for all $y \in \mathbb{R}^n$, then show that $\|A\| \leq \lambda$.
- (iii) Let $A \in L(\mathbb{R}^n, \mathbb{R}^m)$ and $B \in L(\mathbb{R}^m, \mathbb{R}^\ell)$. Show that the product or composition $BA \in L(\mathbb{R}^n, \mathbb{R}^\ell)$ has operator norm $\|BA\| \leq \|B\| \|A\|$.
- (iv) Define $\text{mat}(A)$ or $[A]$ for $A \in L(\mathbb{R}^n, \mathbb{R}^m)$ and call it the *matrix representation* of A w.r.t. the standard bases. Thus $L(\mathbb{R}^n, \mathbb{R}^m)$ is isomorphic to the vector space of $M_{m \times n}(\mathbb{R})$ of $m \times n$ matrices over $\mathbb{R}$. If $\text{mat}(A) = [a_{ij}]_{m \times n}$, then

$$\|A\|_{HS} = \sqrt{\sum_{i=1}^m \sum_{j=1}^n a_{ij}^2}$$

is the Euclidean norm on $M_{m \times n}(\mathbb{R})$, called the *Hilbert-Schmidt norm* of the linear transformation $A \in L(\mathbb{R}^n, \mathbb{R}^m)$. Show that $\|A\| \leq \|A\|_{HS}$.

- (v) Let (X, d) be a metric space and let $a_{ij} : X \rightarrow \mathbb{R}$ be continuous functions for $1 \leq i \leq m$; $1 \leq j \leq n$. If $\Psi : X \rightarrow L(\mathbb{R}^n, \mathbb{R}^m)$ is the map such that

$$\Psi(p) = A_p \in L(\mathbb{R}^n, \mathbb{R}^m) \text{ for } p \in X, \text{ where } \text{mat}(A_p) = [a_{ij}(p)]_{m \times n}.$$

Show that Ψ is continuous.

Q2 Let $E \subseteq \mathbb{R}^n$ be a nonempty open set and $f : E \rightarrow \mathbb{R}^m$ be differentiable at $a \in E$.

- (i) Show that the total derivative $f'(a)$ of f at $a \in E$ is unique. Also, show that f is continuous at a .
- (ii) (Multivariate Chain-Rule) Let $g : G \rightarrow \mathbb{R}^\ell$ be a mapping, G open in $\mathbb{R}^m$, $f(E) \subseteq G$ and $b = f(a) \in G$. If g is differentiable at $b = f(a)$, then show that the composition $g \circ f : E \rightarrow \mathbb{R}^\ell$ is differentiable at a and

$$(g \circ f)'(a) = g'(f(a)) f'(a).$$

- (iii) Suppose that $f(x) = \sum_{i=1}^m f_i(x)u_i$. Show that the partial derivative $(D_j f_i)(a)$ exists and $f'(a) e_j = \sum_{i=1}^m (D_j f_i)(a)u_i$. Deduce that the matrix $\text{mat}(f'(a))$ of the total derivative $f'(a)$ w.r.t. the standard bases is given by

$$\text{mat}(f'(a)) = [(D_j f_i)(a)]_{m \times n},$$

and it is called the *Jacobian matrix* $J_f(a)$ of f at a .

- (iv) Let $\phi: E \rightarrow \mathbb{R}$ be a (real-valued) differentiable function on an open set $E \subseteq \mathbb{R}^n$. For $x \in E$, the *gradient* of ϕ at x is given by $(\nabla \phi)(x) = \sum_{j=1}^n (D_j \phi)(x)e_j$. If $\gamma: (p, q) \rightarrow E$ is a differentiable curve in E and $g = \phi \circ \gamma$ i.e., $g(t) = \phi(\gamma(t))$ for $t \in (p, q)$, then show that

$$g'(t) = \langle (\nabla \phi)(\gamma(t)), \gamma'(t) \rangle,$$

where $\langle \cdot, \cdot \rangle$ is the Euclidean inner product on $\mathbb{R}^n$.

- (v) Let $u \in \mathbb{R}^n$ be a unit vector i.e., $\|u\| = 1$ and ϕ as in (iv). Define the *directional derivative* $(D_u \phi)(x)$ of ϕ at $x \in E$ in the direction of u . Further, show that $(D_u \phi)(x) = \langle (\nabla \phi)(x), u \rangle$. For $\phi(x, y) = \begin{cases} \frac{xy^2}{x^2+y^4} & \text{if } (x, y) \neq (0, 0) \\ 0 & \text{if } (x, y) = (0, 0) \end{cases}$, determine $(D_u \phi)((0, 0))$. Is ϕ continuous at $(0, 0)$? Explain.

Q3 (Mean-Value Theorem)

- (i) Let $\gamma: [0, 1] \rightarrow \mathbb{R}^n$ be a differentiable curve in $\mathbb{R}^n$. Suppose $\|\gamma'(t)\| \leq c$ for all $t \in [0, 1]$ and c is a constant. Then show that $\|\gamma(1) - \gamma(0)\| \leq c$.
- (ii) Let E be a nonempty convex open subset of $\mathbb{R}^n$ and $f: E \rightarrow \mathbb{R}^m$ be a differentiable mapping. Suppose M is a constant and $\|f'(x)\| \leq M$ for all $x \in E$. Then show that $\|f(b) - f(a)\| \leq M \|b - a\|$ for $a, b \in E$. (This is known as a Mean-value theorem for functions of several variables.)
- (iii) Let $\emptyset \neq G \subseteq \mathbb{R}^n$ be a connected open subset and $f: G \rightarrow \mathbb{R}^m$ be a differentiable function. Suppose $f'(x) = 0$ for all $x \in G$. Then show that f is constant on G .

Q4 State and prove Banach contraction principle.